UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II Advanced Calculus (2AC)

Triple integrals

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0.1 Definition and Properties

The construction of a triple integral follows the same process described in the previous chapters for the double. That is, the partition of the domain in finite family of "cubical" boxes, the choice of a sample point in each box in the partition. Then, one writes down the Riemann sums and passes to the limit. As in the case of the double integrals, we are more interested in theses lectures in the evaluation of triple integrals through the successive integration of suitable integrals of functions of a single variable.

0.2 Evaluation over a rectangular box

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a continuous function and let

$$R = [a_1, a_2] \times [b_1, b_2] \times [c_1, c_2] \,.$$

Then

$$\boxed{\iiint_R f(x,y,z)dxdydz = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x,y,z)dzdydx} \tag{1}$$

How many times can one change the order integration in (1)?

0.3 Evaluation over a non-rectangular region

Assume that

$$R = \Big\{(x,y,z) \in \mathbb{R}^3\colon\ (x,y) \in D \subset \mathbb{R}^2 \text{ and } g_1(x,y) \le z \le g_2(x,y)\Big\}$$

where

$$g_1, g_2 : D \to \mathbb{R} \quad \text{are contiunous functions such that } g_1(x,y) \le g_2(x,y) \quad \forall (x,y) \in D \,.$$

So,

$$\boxed{\iiint_R f(x,y,z)dxdydz = \iint_D \left(\int_{g_1(x,y)}^{g_2(x,y)} f(x,y,z)dz \right) dxdy} \tag{2}$$

Now, assume that

$$R = \left\{(x,y,z) \in \mathbb{R}^3 \colon\ z \in [a,b] \text{ and } (x,y) \in R(z)\right\}$$

$R(z)$ is the horizontal section of the region R with the plane passing through the point $(0,0,z)$. In that case,

$$\boxed{\iiint_R f(x,y,z)dxdydz = \int_a^b \left(\iint_{R(z)} f(x,y,z)dxdy\right) dz} \tag{3}$$

Example 0.1 Evaluate the triple integral $\displaystyle\iiint_R (x+z)dxdydz$ where

$$R := \left\{(x,y,z) \in \mathbb{R}^3 \colon\ x \geq 0,\ y \geq 0,\ z \geq 0,\ x+y+z \leq 1\right\}.$$

Solution: We write

$$R = \left\{(x,y,z) \in \mathbb{R}^3 \colon\ (x,y) \in D,\ 0 \leq z \leq 1-x-y\right\}$$

with

$$D = \left\{(x,y) \in \mathbb{R}^2 \colon\ x \geq 0,\ y \geq 0,\ 0 \leq x+y \leq 1\right\}.$$

So,

$$\begin{aligned}
\iiint_R (x+z)dxdydz &= \iint_D \int_0^{1-x-y} (x+z)dzdxdy = \iint_D \left[xz + \frac{1}{2}z^2\right]_0^{1-x-y} dxdy \\
&= \int_0^1 \int_0^{1-x} \left[x(1-x-y) + \frac{1}{2}(1-x-y)^2\right] dydx \\
&= \int_0^1 \left[x(1-x)y - \frac{1}{2}xy^2 - \frac{1}{6}(1-x-y)^3\right]_0^{1-x} dx \\
&= \int_0^1 \left[x(1-x)^2 - \frac{1}{2}x(1-x)^2 + \frac{1}{6}(1-x)^3\right] dx \\
&= \int_0^1 \left[\frac{1}{2}x(1-x)^2 + \frac{1}{6}(1-x)^3\right] dx = \int_0^1 \left[\frac{1}{2}(x-2x^2+x^3) + \frac{1}{6}(1-x)^3\right] dx \\
&= \left[\frac{1}{4}x^2 - \frac{1}{3}x^3 + \frac{1}{8}x^4 - \frac{1}{24}(1-x)^4\right]_0^1 = \frac{1}{4} - \frac{1}{3} + \frac{1}{8} + \frac{1}{24} = \frac{1}{12}
\end{aligned}$$

0.3.1 The change of variables in triple integrals

Theorem 0.2 (Change of variables in triple integrals)
Let $f : R \subset \mathbb{R}^3 \to \mathbb{R}$ be a continuous function. Let $T : R' \subset \mathbb{R}^3 \to \mathbb{R}$ be a function with $(x,y,z) = T(s,t,u)$. Assume that

(i) *T is invertible;*

(ii) *T and T^{-1} are of class $\mathcal{C}^1$ i.e., both functions are differentiable with continuous first order partial derivatives;*

$$\text{(iii)} \quad \boxed{J_T(s,t) := det\left(\frac{\partial(x,y,z)}{\partial(s,t,u)}(s,t,u)\right) := \begin{cases} \begin{vmatrix} x_s(s,t,u) & x_t(s,t,u) & x_u(s,t,u) \\ y_s(s,t,u) & y_t(s,t,u) & y_u(s,t,u) \\ z_s(s,t,u) & z_t(s,t,u) & z_u(s,t,u) \end{vmatrix} \neq 0 \\ \left\{ \begin{array}{l} \textit{except on a finite number} \\ \textit{of surfaces in the region } R'. \end{array} \right. \end{cases}}$$

Then,

$$\boxed{\iiint_R f(x,y,z)dxdydz = \iiint_{R'} f(x(s,t,u),y(s,t,u),z(s,t,u))\big|J_T(s,t,u)\big|dsdtdu} \tag{4}$$

J_T is called the **Jacobian** *of the transformation T.*

The cylindrical coordinates

We reall that we introduced the cylindrical cordinates in the chaper of Parametric surfaces as follows: we take a generic point $M(x,y,z)$ that we project orthogonally onto the xy-plane to obtain the point $A(x,y,0)$. Now, we describe (x,y) in polar coordinates (r,θ) where r is the distance between the point A and the origin O while θ is the angle between the segment line OA and the positive x-axis, by

$$x = r\cos\theta \qquad \text{and} \qquad y = r\sin\theta.$$

We then obtain the set of coordinates (r,θ,z) called the cylindrical coordinates of the point M and defined through the equations:

$$\boxed{x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z}.$$

We get a tranformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ who Jacobian is

$$\boxed{J_T(r,\theta,z) := \begin{vmatrix} x_r(r,\theta,z) & x_\theta(r,\theta,z) & x_z(r,\theta,z) \\ y_r(r,\theta,z) & y_\theta(r,\theta,z) & y_z(r,\theta,z) \\ z_r(r,\theta,z) & z_\theta(r,\theta,u) & z_z(r,\theta,z) \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r}. \tag{5}$$

Spherical coordinates:

For the description of the spherical coordinates, we take a generic point $M(x,y,z)$ and consider the distance ρ between the point M and the origin O ($\rho := \sqrt{x^2+y^2+z^2}$) and the angle ϕ between the segment line OM and the positive z-axis. Now, if we project the point M orthogonal onto the z-axis, to get $z = \rho\cos\phi$. On the other hand, if we project orthogonally the point M onto the xy-plane we obtain the point $A(x,y,0)$. Now, we describe (x,y) in polar coordinates as above with the difference that we use the triangle OMA in order to express r in terms of ρ and ϕ and get that $r = \rho\sin\phi$. Therefore, we obtain the set of coordinates (ρ,θ,ϕ) called the spherical coordinates of the point M and defined through the equations:

$$\boxed{x = \rho\cos\theta\sin\phi, \qquad y = \rho\sin\theta\sin\phi, \qquad z = \rho\cos\phi}.$$

We get a tranformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ who Jacobian is

$$\boxed{\begin{aligned} J_T(\rho,\theta,\phi) &= \begin{vmatrix} x_\rho(\rho,\theta,\phi) & x_\theta(\rho,\theta,\phi) & x_\phi(\rho,\theta,\phi) \\ y_\rho(\rho,\theta,\phi) & y_\theta(\rho,\theta,\phi) & y_\phi(\rho,\theta,\phi) \\ z_\rho(\rho,\theta,\phi) & z_\theta(\rho,\theta,\phi) & z_\phi(\rho,\theta,\phi) \end{vmatrix} \\ &= \begin{vmatrix} \cos\theta\sin\phi & -\rho\sin\theta\sin\phi & \rho\cos\theta\cos\phi \\ \sin\theta\sin\phi & \rho\cos\theta\sin\phi & \rho\sin\theta\cos\phi \\ \cos\phi & 0 & -\rho\sin\phi \end{vmatrix} = -\rho^2\sin\phi \end{aligned}} . \tag{6}$$

Evaluation of triple integrals through cylindrical and spherical coordinates

Example 0.3

(a) Evaluate the triple integral $\iiint_R 2z dx dy dz$ where

$$R := \left\{(x,y,z) \in \mathbb{R}^3 \colon\ x^2+y^2 \le 1,\ 0 \le z \le 3\sqrt{x^2+y^2}\right\}.$$

Using cylindrical coordinates we find that R is the image of

$$R' := \left\{(r,\theta,z) \in \mathbb{R}^3 \colon\ 0 \le r \le 1,\ 0 \le \theta \le 2\pi,\ 0 \le z \le 3r\right\}.$$

with Jacobian equal to r so that

$$\begin{aligned} \iiint_R 2z dx dy dz &= \int_0^1 \int_0^{2\pi} \int_0^{3r} 2zr dz d\theta dr = \int_0^1 \int_0^{2\pi} \left[z^2 r\right]_0^{3r} d\theta dr \\ &= \int_0^1 \int_0^{2\pi} 9r^3 d\theta dr = 2\pi\left[\frac{9}{4}r^4\right]_0^1 = \frac{9\pi}{2}. \end{aligned}$$

(b) Evaluate the triple integral $\iiint_R (x^2+y^2+z-2) dx dy dz$ where R is the quarter of the sphere centered at the origin with radius 2, which is in the region $x \ge$ and $y \ge 0$. Hint: Use spherical coordinates.

(c) Evaluate the triple integral $\iiint_R 2x dx dy dz$ where

$$R := \left\{(x,y,z) \in \mathbb{R}^3 \colon\ x \ge 0,\ 0 \le z \le 2y+1,\ x^2+4y^2+z^2 \le 1\right\}.$$

Hint: You can write $R := \left\{(x,y,z) \in \mathbb{R}^3 \colon\ (y,z) \in D \text{ and } 0 \le x \le \sqrt{1-4y^2-z^2}\right\}$ with

$$D := \left\{(y,z) \in \mathbb{R}^2 \colon\ 0 \le z \le 2y+1,\ 4y^2+z^2 \le 1\right\}$$

$$\iiint_R 2x dx dy dz = \iint_D \left(\int_0^{\sqrt{1-4y^2-z^2}} 2x dx\right) dy dz$$

Exercises 0.4 Evaluate the triple integral in the corresponding region.

(a) $\iiint_R (x^2 + yz)dxdydz$ where R is the region inside the sphere centered at the origin with radius 1 and above the cone $z = \sqrt{x^2 + y^2}$. Answer: $\boxed{\frac{\pi}{15}\left(2 - \frac{5\sqrt{2}}{4}\right)}$

(b) $\iiint_R (x^2 + y^2 + z - 2))dxdydz$ where R is the quarter of the sphere centered at the origin $(0, 0, 0)$ with radius 2 found the the region $x \geq 0,\ y \geq 0$. (Hint: Use spherical coordinates!)

We will start vector calculus with the study of vector fields in the next set of notes!